

HARSHITHA SAI MADDUKURI

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Work Authorization: CPT/OPT (F-1) · Expected Graduation: May 2027

Summary

Full Stack Software Engineer with hands-on experience building production web and mobile applications using TypeScript, React, React Native, Node.js, Express.js, Tailwind CSS, and SQL. Comfortable working across the full stack — from designing relational database schemas and REST APIs to building responsive, type-safe frontends. Fast learner who thrives in collaborative, innovation-driven environments and enjoys shipping features that real users depend on.

Education

Northeastern University, Boston, Master of Science in Computer Science, Khoury College **Jan 2025 – May 2027**

Courses: Algorithms, Data Structures, Machine Learning, Database Management Systems, Web Development, Cyber Security

Vasireddy Venkatadri Institute of Technology, India, B.Tech in Computer Science and Engineering **2019 – 2023**

Courses: OOP, Computer Networks, Operating Systems, Software Engineering, Statistics

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C#, SQL

Frontend & Mobile: React, React Native, Next.js, Tailwind CSS, Redux, HTML5, CSS3, Bootstrap

Backend & APIs: Node.js, Express.js, REST APIs, Session-Based Authentication

Database & Tools: MongoDB, SQL Databases, Mongoose, Git/GitHub, Vercel, AWS, CI/CD

Practices: Agile/Scrum, Debugging, Unit Testing, Code Reviews, Technical Documentation

Work Experience

Software Engineer Intern — Virtusa **Dec 2023 – May 2024**

- Completed Virtusa's structured internal training program covering full-stack development and enterprise best practices; passed formal technical assessments to qualify for production project work.
- Developed production-ready full-stack features end-to-end using .NET (C#) backend and React/TypeScript frontend, taking ownership from requirements through deployment across Agile sprints.
- Built RESTful APIs with JWT-based authentication and role-based access control; wrote and optimized SQL queries and relational schemas to manage live production data reliably.
- Debugged issues across frontend, API, and database layers systematically; authored technical documentation and contributed to code reviews to maintain code quality across the team.
- Collaborated with cross-functional stakeholders to translate business requirements into well-scoped technical solutions, consistently communicating progress and iterating quickly on feedback.

Projects

Kambaz Learning Management System | Full-Stack Web Application **2025**

- Built a full-stack LMS across 6 progressive modules using Next.js, React, Redux, and TypeScript on the frontend with Tailwind CSS for styling and TanStack-style data-fetching patterns for efficient server state management.
- Developed a complete Node.js/Express.js backend with modular routes and DAOs for courses, modules, users, and enrollments; integrated MongoDB Atlas with Mongoose schemas and full CRUD operations across all resources.
- Implemented session-based authentication, role-based access control (student/instructor/admin), and many-to-many enrollment relationships using a dedicated enrollments collection with Mongoose populate().
- Deployed frontend to Vercel and backend to Render.com; configured environment variables across local and production environments with Git-based branch workflows. **Tech:** Next.js, React, TypeScript, Tailwind CSS, Redux, Node.js, Express.js, MongoDB, Mongoose, SQL, Vercel, Git

Signature Verification using Siamese Neural Networks | Published Research **2023**

- Reproduced a Siamese CNN architecture in Python for offline signature verification — built the complete pipeline covering preprocessing, contrastive loss-based training, and threshold-based classification of genuine vs. forged pairs.
- Independently authored and published a full research paper in IJFANS Journal covering methodology, results, and conclusions, demonstrating strong technical writing and analytical communication. **Tech:** Python, Java

Self-Balancing Binary Search Trees | CS5800 Algorithms, Northeastern **2025**

- Implemented AVL and Red-Black Trees in Python from scratch with full CRUD, traversals, and a comprehensive automated test suite covering all edge cases including empty trees, duplicates, and large inputs.
- Benchmarked 1K–10K node datasets on random, sorted, and reverse-sorted inputs; authored a structured written analysis documenting $O(\log n)$ vs. $O(n)$ trade-offs and AVL vs. RBT behavioral differences. **Tech:** Python, LaTeX, Excel

Additional

Certifications: IBM Full Stack Developer, Microsoft AZ-900 **Languages:** English, Telugu, Hindi